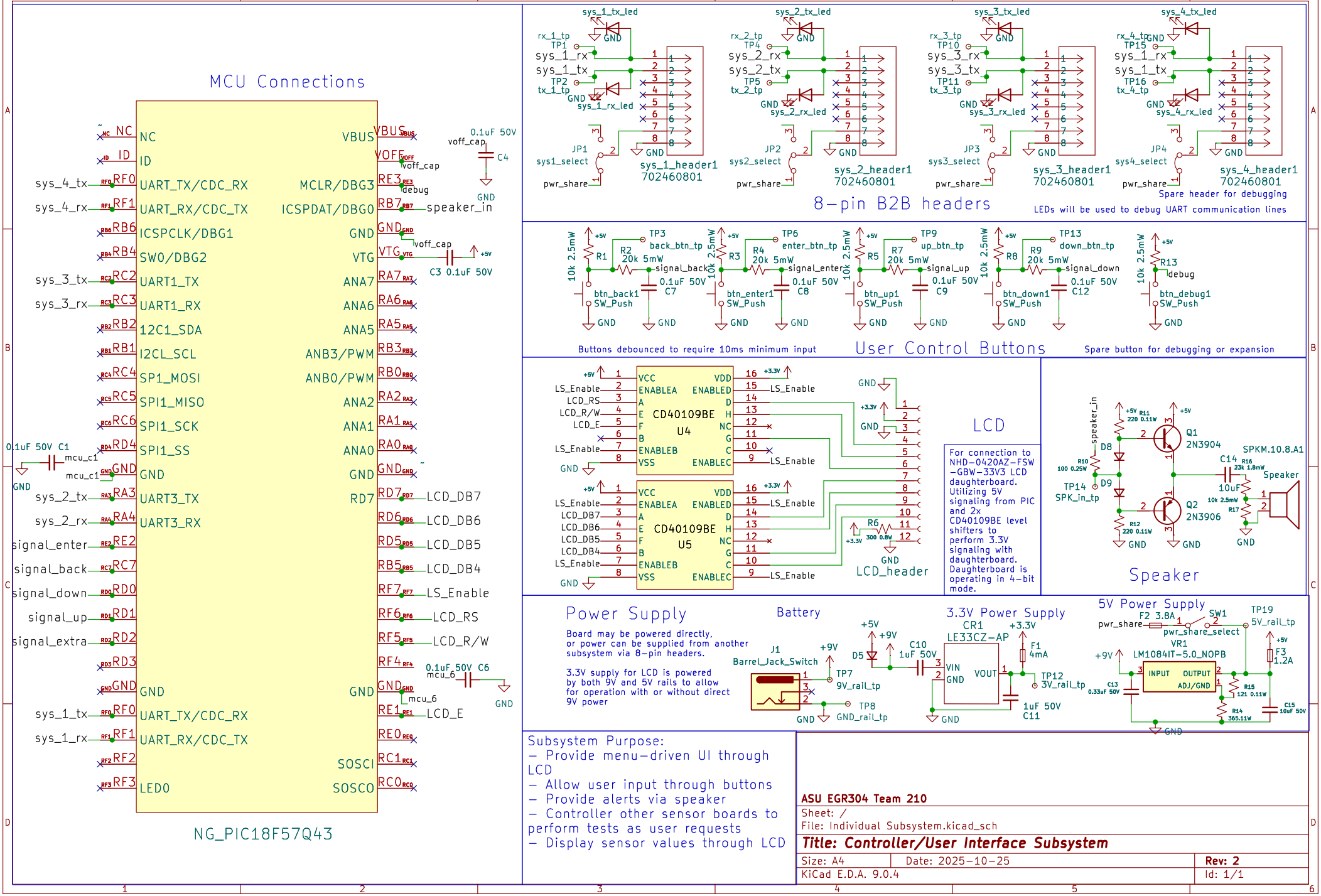
MCU Connections
NG_PIC18F57Q43
8-pin B2B headers
sys_1_header1 702460801
sys_2_header1 702460801
sys_3_header1 702460801
sys_4_header1 702460801
Spare header for debugging
LEDs will be used to debug UART communication lines
User Control Buttons
Buttons debounced to require 10ms minimum input
Spare button for debugging or expansion
CD40109BE U4
CD40109BE U5
LCD_header
LCD
For connection to NHD-0420AZ-FSW-GBW-33V3 LCD daughterboard. Utilizing 5V signaling from PIC and 2x CD40109BE level shifters to perform 3.3V signaling with daughterboard. Daughterboard is operating in 4-bit mode.
Speaker
Power Supply
Board may be powered directly, or power can be supplied from another subsystem via 8-pin headers.
3.3V supply for LCD is powered by both 9V and 5V rails to allow for operation with or without direct 9V power
Battery
3.3V Power Supply
5V Power Supply
Subsystem Purpose:
- Provide menu-driven UI through LCD
- Allow user input through buttons
- Provide alerts via speaker
- Controller other sensor boards to perform tests as user requests
- Display sensor values through LCD
ASU EGR304 Team 210
Sheet: /
File: Individual Subsystem.kicad_sch
Title: Controller/User Interface Subsystem
Size: A4
Date: 2025-10-25
Rev: 2
KiCad E.D.A. 9.0.4
Id: 1/1